

HapCap: Estimating Haptic Adjectives with Vision-Language Model for Dataset Annotation

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Abstract. Haptic datasets capturing human haptic perceptual ratings underpin haptic texture rendering and cross-modal understanding. However, such datasets are scarce and hard to scale, because annotating objects with human perceptual ratings requires participants to physically handle each item. On the other hand, human haptic perception is closely tied to an object’s visual appearance and linguistic description, suggesting that tactile attributes may be inferable without physical contact. We propose HapCap, a haptic-adjective annotation method using Vision-Language Model (VLM) trained on large-scale image-text corpora. We feed each object’s photograph and name from the dataset into the model to estimate adjective ratings. To investigate how well VLMs can reproduce human haptic judgments, we benchmark three open-weight models on a haptic dataset with human perceptual ratings. All three models stayed within roughly one point of human ratings on the 1–5 scale on average. They broadly captured human haptic perception. These preliminary results position VLMs as a promising alternative that can substantially reduce the cost of haptic dataset annotation.

Keywords: Haptic dataset annotation · Vision-Language Model.

1 Introduction

Datasets annotated with human haptic perceptual ratings underpin haptic texture rendering and emerging cross-modal systems that jointly reason about vision, language, and touch. Building such datasets requires multiple participants to physically handle each object and rate how it feels on haptic adjectives such as soft, rough, or warm. Collecting these ratings at scale is costly, and datasets of this kind remain particularly scarce [1,3].

On the other hand, human haptic perception is closely linked to visual and semantic cues. Given only a photograph or an object name, people often guess its temperature, hardness, or roughness with reasonable accuracy [2]. This suggests that a model trained on such cues could produce haptic adjective ratings comparable to human ratings, even without such costly physical handling.

We propose HapCap, a haptic-adjective annotation method built on Vision-Language Model (VLM) that takes each object’s photograph and name from

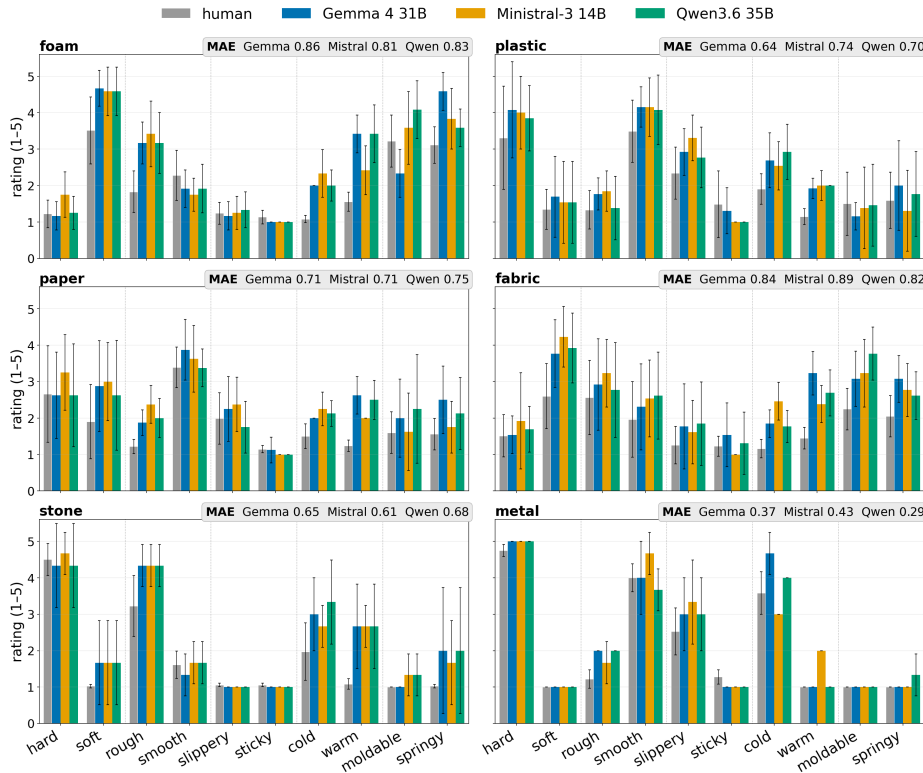


Fig. 1. Rating (1–5 scale) per material category and adjective for all three VLMs and the human baseline, with per-model MAE annotated at the top of each panel. Error bars are SD across objects within the category.

the dataset as input. VLMs are trained on large-scale image-text corpora and have been shown to capture broad knowledge about how everyday objects look and how people describe them, making them natural candidates for mapping visual and linguistic cues to tactile adjectives. In this paper, to evaluate HapCap, we compare its predicted ratings against human ratings on a publicly available haptic dataset [3] that provides per-object rating distributions on ten tactile adjectives across 60 household objects, along with representative photographs. A category-level comparison shows that the best-performing model closely tracks human ratings on most material-adjective cells, positioning VLMs as a promising route to substantially reduce the cost of haptic dataset annotation.

2 Method

HapCap frames haptic adjective annotation as a zero-shot VLM query: given an object photograph and its name, the model returns an integer score on a fixed scale for each adjective in a predefined set. Concretely, the input is delivered as a single prompt: a system message assigns the role of a haptic perception expert, and the user message provides the object’s photograph and name. The output

is constrained to strict JSON containing one integer score and a short visual rationale per adjective.

3 Experiment

We evaluate how well HapCap reproduces human haptic adjective ratings across different VLM backends. We use objects from the Penn Haptic Adjective Corpus 2 (PHAC-2) [3] that have both a representative photograph and human 1–5 ratings on the ten haptic adjectives. We evaluate three open-weight VLMs from three vendors, served locally through Ollama on a single RTX 5090 (32 GB VRAM): Qwen3.6 35B, Ministral-3 14B, and Gemma 4 31B. We group objects into six material categories and compare the category-wise mean rating of each VLM against the human mean on every adjective (Fig. 1).

Across the six panels, the three VLMs largely reproduce the shape of the human profile, with bars rising and falling together within each category. The mean MAE against human ratings stays below one point on the 1–5 scale for all three models (0.74 for Gemma 4 and Qwen3.6, 0.76 for Ministral-3). There are no substantial differences among them.

Per-category MAE ranges from 0.36 on metal to 0.85 on fabric, with foam (0.83) being a close second. On foam and fabric in particular, the models miss most on warm and soft, overestimating both by about 1.0–1.5 points. This likely reflects the models’ reliance on visual and textual associations rather than on actual thermal or compliance cues, which are inaccessible from a photograph.

4 Conclusion, Open Research Questions, and Interdisciplinary Exchange

This work proposed HapCap, a haptic-adjective annotation method that takes each object’s photograph and name as input to a VLM. On the PHAC-2 dataset, three open-weight VLMs reproduced human ratings within a mean MAE of about one point on the 1–5 scale. They tracked the shape of the human profile in most material categories. However, warm and soft were consistently overestimated for fabric and foam.

Future work will extend HapCap by feeding haptic contact signals from the dataset as additional input. We will also statistically examine whether VLM ratings agree with human ratings. From a machine-learning perspective, training-free techniques to enhance the inference ability of existing VLMs are worth exploring.

References

1. Balasubramanian, J.K., Kodak, B.L., Vardar, Y.: Sens3: Multisensory database of finger-surface interactions and corresponding sensations. In: EuroHaptics 2024. p. 262–277 (2024)
2. Baumgartner, E., Wiebel, C.B., Gegenfurtner, K.R.: Visual and haptic representations of material properties. *Multisensory Research* **26**(5), 429–455 (2013)
3. Chu, V., McMahon, I., Riano, L., McDonald, C.G., He, Q., Martinez Perez-Tejada, J., Arrigo, M., Darrell, T., Kuchenbecker, K.J.: Robotic learning of haptic adjectives through physical interaction. *Robot. Auton. Syst.* **63**(P3), 279–292 (2015)